A1 Week 2 Plan

Role 1 (Optimal Contract and General Carbon Theory)

- **Prove existence, uniqueness, and contract optimality.** Prove expected profit is coercive and, under a primitive condition on F (e.g. a log-concave density), strictly concave on the interior region, so the capacity optimum is unique; and prove the dispatch contract d*(x)=min{(K_F−x)_+, K_M} is the unique optimal policy via the strict R>c ordering. This supplies the negative-definite-Hessian condition from primitives instead of assuming it.
- **Promote the shift-ratio condition to a primitive.** Prove that a support-wide monotonicity of g(x)=F_{K_R}/f (or a hazard-rate condition on F) implies the endogenous-window condition the capacity channel currently relies on; give the implication with a proof.
- **Prove the necessary-and-sufficient emission condition for general F.** Under that primitive plus a tail/skew condition, prove a sharp iff for carbon reduction at a finite interior optimum for any such F, upgrading the endogenous inequality (C) and the existence conjecture to a theorem in primitives. Exhibit a non-power-law F satisfying all conditions to prove the class non-vacuous.
- **Prove single-crossing and the viability threshold for general F.** Extend the renewable-cost window from the power-law family to general F and prove $\Delta(C_R)$ is single-crossing, so the carbon-reduction set is the interval {C_R < C_R†} and the strict fall occurs at a finite interior K_R* rather than the unbounded corner.
- **Prove sharpness and tight magnitude bounds.** Prove the shift-ratio condition is necessary as well as sufficient for K_R* > K_R° (a realized boundary, not just an algebraic sign), and prove the two-sided bounds on the carbon saving Δ are tight, characterizing the carbon-reduction region in primitive parameter space.

Role 2 (Solver and Numerical Methods)

- **Build and validate the general solver.** Implement the joint-capacity and dispatch solver for d*(x) and (K_R,K_F,K_M) under any residual-demand distribution, and validate it against the closed forms to machine precision — the verified backend the rest of the group computes against.
- **Run the carbon-sign sweep at a finite optimum.** Sweep over Role 1's primitive condition on a curated distribution family and show, numerically, that carbon falls at a finite interior optimum — replacing the scattered Week-1 counts with one principled map.
- **Certify the Hessian on the interior path.** Numerically certify the profit Hessian is negative definite along the interior equilibrium path for Role 1's distribution class, by eigenvalue computation, backstopping existence and uniqueness.
- **Derive and implement the analytic Hessian and cofactors.** Derive the full analytic Hessian and the comparative-statics cofactors (replacing finite differences, which are noisy at the knife-edge) and certify them against the solved equilibria across the class.

Role 3 (Comparative Statics and Robustness)

- **Prove the correlation and price comparative statics.** Prove the net-carbon sign is governed by the demand–supply association g = E[S | x]; that mining raises optimal

renewable capacity $K_R^* > K_R^\circ$; and sign ∂(carbon effect)/∂R, $\partial/\partial\beta$, and $\partial/\partial p$ via the Hessian-cofactor identities, under general conditions.

- **Prove the cross-distribution robust certificate.** Sign the capacity covariation dK^*/dF along the shape class U so the least-favorable result holds across F — making $C_R \le C_R^\dagger \Rightarrow \Delta \ge 0$ a genuine distributionally-robust certificate — or prove the obstruction.
- **Construct the reversed-channel counterexample.** Build a realized distribution F with g increasing for which $K_R^* < K_R^\circ$, demonstrating the channel reverses — the sharpness example Week 1 could not produce — and solve it numerically to confirm.
- **Prove the limit-interchange rigorously.** Replace the expected-fossil-to-zero corner step with a dominated-convergence or finite-capacity envelope argument, so the threshold proof holds without the a.s.-limit shortcut.

Role 4 (Mechanism Design and Regimes)

- **Prove the principal–agent implementation with IR and IC.** Cast the regulator as principal and the utility/miner as agent; write the participation (IR) and incentive-compatibility (IC) constraints and prove the price-based program implements d^* optimally and, where feasible, uniquely — not merely as the principal's first-best dispatch — stating the price-taking / exogenous-R conditions under which IR and IC are slack.
- **Prove the unified regime theorem.** Combine the C_M coverage axis (no adoption / small-FCL / interior / full coverage) and the C_R boundedness axis into one theorem, and pin down the small-FCL crossover $K_M^* = K_R^\circ$ that Week 1 left open.
- **Prove the necessity of Assumption 1.** For each of the four parameter conditions, prove it is necessary for the interior optimum or the economic sign it guards, and characterize the regime obtained when it is relaxed (full-coverage corner, no adoption, or a perverse curtailment).
- **Characterize the second-best contract under private information.** When the agent privately observes its availability $a(x)=(K_F-x)_+$, derive the optimal screening contract (the second-best d^*) and characterize how the resulting information rent changes the emission outcome versus the full-information benchmark.